EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 11A

1. From LCCDE to State Space: A Second-Order Recursive Filter

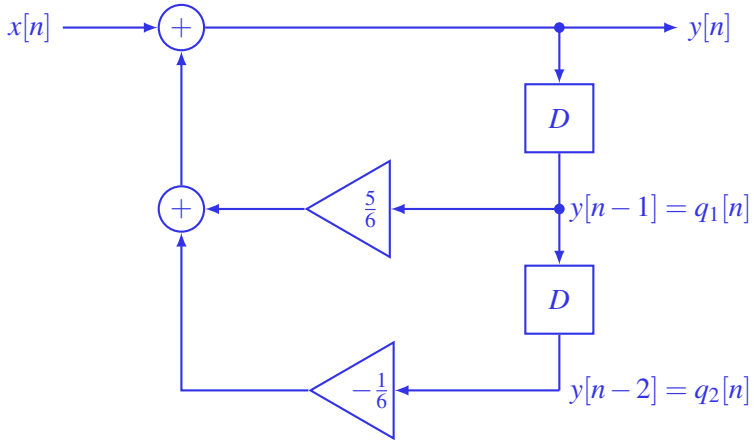
Consider the causal discrete-time LTI system with input $x[n]$ and output $y[n]$ described by the linear constant-coefficient difference equation (LCCDE)

$$\forall n \in \mathbb{Z}, \quad y[n] = \frac{5}{6}y[n-1] - \frac{1}{6}y[n-2] + x[n].$$

Assume the system is initially at rest: $y[n] = 0$ for all $n < 0$.

(a) Draw a delay-adder-gain (DAG) block diagram that implements this LCCDE. Use only delay blocks (D), adders ($+$), and gain blocks. Label the outputs of your delay blocks.

Answer: The output $y[n]$ is the sum of $x[n]$ and scaled copies of past outputs. Passing $y[n]$ through one delay gives $y[n-1]$, and passing that through a second delay gives $y[n-2]$. We scale these by $\frac{5}{6}$ and $-\frac{1}{6}$ and feed them back into the adder:



(b) We now write this system in state-space form. Choose the outputs of the two delay blocks as the state variables:

$$\mathbf{q}[n] = \begin{bmatrix} q_1[n] \\ q_2[n] \end{bmatrix} = \begin{bmatrix} y[n-1] \\ y[n-2] \end{bmatrix}.$$

Write scalar equations for $q_1[n+1]$, $q_2[n+1]$, and $y[n]$ in terms of $q_1[n]$, $q_2[n]$, and $x[n]$.

Answer: Substituting the state variables into the LCCDE gives the output equation:

$$y[n] = \frac{5}{6}q_1[n] - \frac{1}{6}q_2[n] + x[n].$$

The first delay block has input $y[n]$, so its output at the next time step is

$$q_1[n+1] = y[n] = \frac{5}{6}q_1[n] - \frac{1}{6}q_2[n] + x[n].$$

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The second delay block has input $y[n-1] = q_1[n]$, so

$$q_2[n+1] = y[n-1] = q_1[n].$$

(c) Express the system in matrix state-space form

$$\mathbf{q}[n+1] = A\mathbf{q}[n] + Bx[n], \qquad y[n] = C\mathbf{q}[n] + Dx[n].$$

Find A, B, C, D, and the initial state $\mathbf{q}[0]$.

Answer: Reading off the coefficients from the scalar equations in the previous part:

$$A = \begin{bmatrix} \frac{5}{6} & -\frac{1}{6} \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} \frac{5}{6} & -\frac{1}{6} \end{bmatrix}, \quad D = 1.$$

Because the system is initially at rest, $y[-1] = y[-2] = 0$, so

$$\mathbf{q}[0] = \begin{bmatrix} y[-1] \\ y[-2] \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

(d) The impulse response $h[n]$ is the output $y[n]$ when the input is $x[n] = \delta[n]$ and the system starts at rest. Use your state-space equations to compute $h[0]$, $h[1]$, $h[2]$, and $h[3]$.

Answer: We start from $\mathbf{q}[0] = \mathbf{0}$, with $x[0] = 1$ and $x[n] = 0$ for $n \geq 1$.

$n = 0$**:** $h[0] = C\mathbf{q}[0] + D \cdot 1 = 1$, and $\mathbf{q}[1] = A\mathbf{0} + B \cdot 1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

$n = 1$**:** $h[1] = C\mathbf{q}[1] = \frac{5}{6}$, and $\mathbf{q}[2] = A\mathbf{q}[1] = \begin{bmatrix} \frac{5}{6} \\ 1 \end{bmatrix}$.

$n = 2$**:** $h[2] = C\mathbf{q}[2] = \frac{5}{6} \cdot \frac{5}{6} - \frac{1}{6} \cdot 1 = \frac{25}{36} - \frac{6}{36} = \frac{19}{36}$, and $\mathbf{q}[3] = A\mathbf{q}[2] = \begin{bmatrix} \frac{19}{36} \\ \frac{5}{6} \end{bmatrix}$.

$n = 3$**:** $h[3] = C\mathbf{q}[3] = \frac{5}{6} \cdot \frac{19}{36} - \frac{1}{6} \cdot \frac{5}{6} = \frac{95}{216} - \frac{30}{216} = \frac{65}{216}$.

So $h[0] = 1$, $h[1] = \frac{5}{6}$, $h[2] = \frac{19}{36}$, and $h[3] = \frac{65}{216}$.

(e) Now find a closed-form expression for $h[n]$.

i. Show that $h[0] = D$ and that $h[n] = CA^{n-1}B$ for $n \geq 1$.

ii. You are given that A has eigenvalues $\lambda_1 = \frac{1}{2}$ and $\lambda_2 = \frac{1}{3}$, with corresponding eigenvectors

$$\mathbf{v}_1 = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} \frac{1}{3} \\ 1 \end{bmatrix}.$$

Diagonalize $A = V\Lambda V^{-1}$ and use this to find $h[n]$ for all n. Check your answer against the values from the previous part.

Answer: **(i)** With $x[n] = \delta[n]$ and $\mathbf{q}[0] = \mathbf{0}$, we get $h[0] = C\mathbf{0} + D = D$ and $\mathbf{q}[1] = B$. For $n \geq 1$ the input is zero, so $\mathbf{q}[n+1] = A\mathbf{q}[n]$, which gives $\mathbf{q}[n] = A^{n-1}\mathbf{q}[1] = A^{n-1}B$. Therefore

$$h[n] = C\mathbf{q}[n] + D \cdot 0 = CA^{n-1}B, \qquad n \geq 1.$$

(ii) Stack the eigenvectors into V and the eigenvalues into Λ:

$$V = \begin{bmatrix} \frac{1}{2} & \frac{1}{3} \\ 1 & 1 \end{bmatrix}, \qquad \Lambda = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{bmatrix}, \qquad V^{-1} = \frac{1}{\frac{1}{2} - \frac{1}{3}} \begin{bmatrix} 1 & -\frac{1}{3} \\ -1 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 6 & -2 \\ -6 & 3 \end{bmatrix}.$$

Then $A^{n-1} = V\Lambda^{n-1}V^{-1}$, so $h[n] = (CV)\,\Lambda^{n-1}\,(V^{-1}B)$. Computing the two outer factors:

$$CV = \begin{bmatrix} \frac{5}{6}\cdot\frac{1}{2} - \frac{1}{6} & \frac{5}{6}\cdot\frac{1}{3} - \frac{1}{6} \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & \frac{1}{9} \end{bmatrix}, \qquad V^{-1}B = \begin{bmatrix} 6 \\ -6 \end{bmatrix}.$$

So for $n \geq 1$,

$$h[n] = \frac{1}{4}\cdot 6\left(\frac{1}{2}\right)^{n-1} - \frac{1}{9}\cdot 6\left(\frac{1}{3}\right)^{n-1} = 3\left(\frac{1}{2}\right)^{n} - 2\left(\frac{1}{3}\right)^{n}.$$

At $n = 0$ this formula gives $3 - 2 = 1 = D$, so it holds for $n = 0$ as well. Since $h[n] = 0$ for $n < 0$,

$$\boxed{h[n] = \left(3\left(\frac{1}{2}\right)^{n} - 2\left(\frac{1}{3}\right)^{n}\right)u[n].}$$

Check: $h[1] = \frac{3}{2} - \frac{2}{3} = \frac{5}{6}$, $h[2] = \frac{3}{4} - \frac{2}{9} = \frac{19}{36}$, and $h[3] = \frac{3}{8} - \frac{2}{27} = \frac{81-16}{216} = \frac{65}{216}$. These match the previous part.

(f) Is this system BIBO stable? Justify your answer using the impulse response. How does your answer relate to the eigenvalues of A?

Answer: A discrete-time LTI system is BIBO stable if and only if its impulse response is absolutely summable, $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$. By the triangle inequality and the geometric series formula,

$$\sum_{n=0}^{\infty} |h[n]| \leq 3\sum_{n=0}^{\infty}\left(\frac{1}{2}\right)^{n} + 2\sum_{n=0}^{\infty}\left(\frac{1}{3}\right)^{n} = 3\cdot\frac{1}{1-\frac{1}{2}} + 2\cdot\frac{1}{1-\frac{1}{3}} = 6 + 3 = 9 < \infty.$$

So the system **is BIBO stable**.

Each term of $h[n]$ is a power of one of the eigenvalues of A, because $h[n] = CV\Lambda^{n-1}V^{-1}B$. Both eigenvalues satisfy $|\lambda_k| < 1$, so both geometric series converge. If A had an eigenvalue with $|\lambda| \geq 1$ whose term appeared in $h[n]$, then $h[n]$ would not decay to zero, the sum would diverge, and the system would not be BIBO stable.